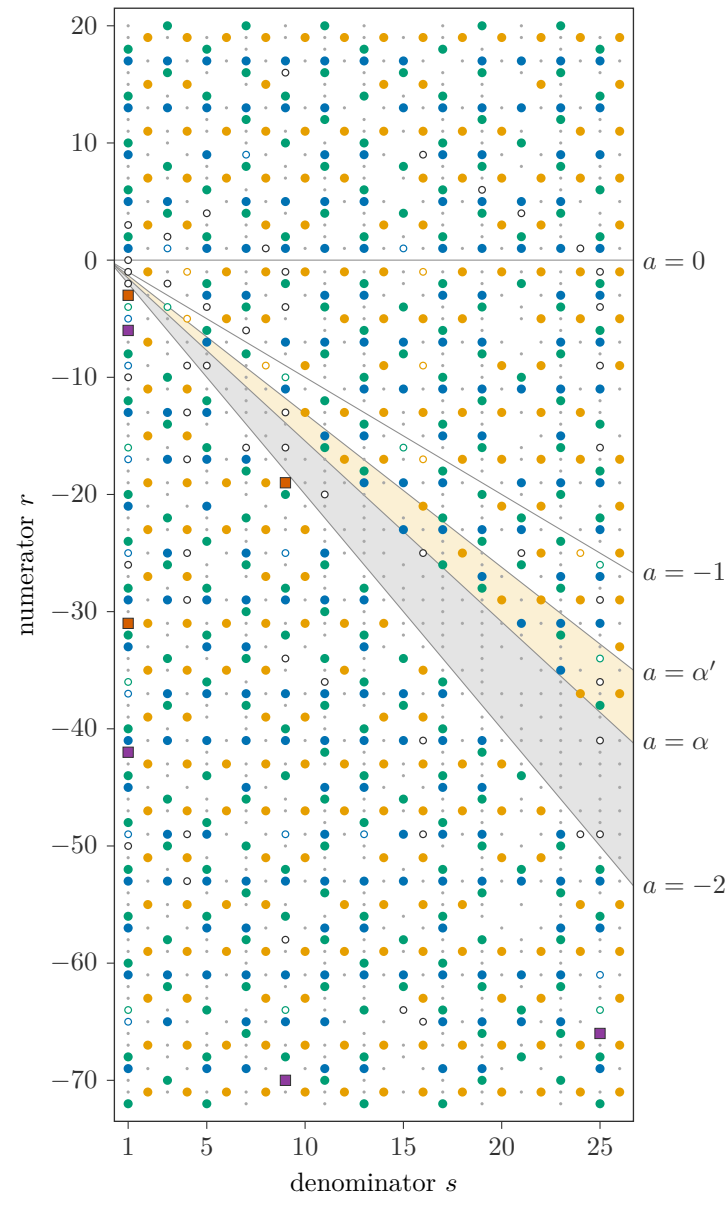


coprime r, s with $s \geq 1$, $a = r/s$

Main theorem: circles

(r, s) mod 4 for $a \leq -2$; for $a > \alpha$

- (C1): $(3,1)$, $(3,3)$; $(1,1)$, $(1,3)$
- (C2): $(0,1)$, $(2,3)$; $(2,1)$, $(0,3)$
- (C3): $(1,0)$, $(1,2)$; $(3,0)$, $(3,2)$

two of (Z), (O), (T) hold, so ρ_a is surjective

c_1, c_2 are 2-dependent, so ρ_a is not surjective

The two families: squares

- $a = -\ell(\ell+1) \leq -2$, $\ell = j/k$, $(j,k) \equiv (2,1), (3,3) \bmod 4$
- $a = -(\ell^2+\ell+1) \leq -2$, $\ell = j/k$, $(j,k) \equiv (1,1), (2,3) \bmod 4$

Not covered

$c_1, \ldots, c_6$ are 2-dependent, so ρ_a is not surjective

$c_1, \ldots, c_6$ are 2-independent

no mark: $\gcd(r,s) > 1$; a appears at its reduced fraction, whose residues mod 4 may differ

Regions

$\alpha \approx -1.544$, $\alpha' \approx -1.311$

$-2 < a \leq \alpha$: not covered

$\alpha < a \leq \alpha'$; outside it, in the classes, ρ_a is surjective iff c_1, c_2 are 2-independent